

Project Update: Dynamics and Stability of the Spring-Loaded Inverted Pendulum (SLIP) Model

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(a) System and Motivation

For my final project I am modeling the spring-loaded inverted pendulum (SLIP) model of a hopping/running robot – the reduced-order model that underlies essentially all legged-robot dynamics. Despite its simplicity (a point mass bouncing on a massless spring leg), the SLIP model captures the center-of-mass dynamics of running in both animals and machines remarkably well, making it simple enough to analyze rigorously with the tools from this course while still producing rich, interesting behavior. It also builds naturally on the elastic pendulum system studied in class, since the stance phase is essentially an inverted spring pendulum.

(b) Modeling Methods

The SLIP model is a hybrid dynamical system, shown schematically in Figure 1: during stance the body behaves as a spring-loaded pendulum, while during flight it is simply a projectile. I am deriving the equations of motion for each phase using the Lagrangian formulation in polar coordinates (r, θ) for the stance phase, together with the switching conditions at touchdown and liftoff that connect the two phases. To assess stability of the hopping gait itself (rather than just a single equilibrium), I will use an apex return map – a Poincaré map in which a periodic hopping gait appears as a fixed point whose stability can be evaluated numerically [1]. The model will be implemented in Python using event-based numerical integration, with a parameter study over spring stiffness and touchdown angle.

(c) Progress to Date

TODO: Summarize the progress made so far – e.g., which equations of motion have been derived, whether the touchdown/liftoff switching conditions are implemented, whether the Python simulation runs, and any preliminary results (e.g., a first stance-phase trajectory or return-map point).

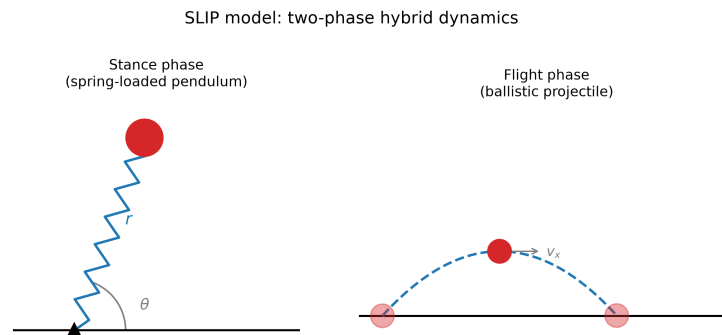


Figure 1: Schematic of the SLIP model's two-phase hybrid dynamics: stance (spring-loaded pendulum in polar coordinates r, θ) and flight (ballistic projectile).

Scheduled update meeting: TODO – day, date, and time.

References

- [1] R. Blickhan, “The spring-mass model for running and hopping,” *Journal of Biomechanics*, vol. 22, no. 11–12, pp. 1217–1227, 1989.